



D.4 Electromagnetic induction (HL)

Fields · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Magnetic flux through a surface is $\Phi = BA \cdot \cos\theta$, where θ is the angle between the field B and the normal to the area A . Faraday's law states that the electromotive force (emf) induced in a circuit equals the negative rate of change of magnetic flux linkage through it, $\epsilon = -N(\Delta\Phi/\Delta t)$ for N turns, and Lenz's law (embodied in the minus sign) states that the induced emf/current always acts to oppose the change in flux that produced it, consistent with conservation of energy. In an AC generator, a coil rotating at constant angular velocity ω in a uniform field produces a sinusoidally varying emf, $\epsilon = \epsilon_0 \sin(\omega t)$, where $\epsilon_0 = NBA\omega$ is the peak emf; transformers use this changing flux (via mutual induction) so that for an ideal transformer, $V_s/V_p = N_s/N_p$, with power conserved ($V_{pl_p} = V_{sl_s}$).

Worked example: follow the reasoning

A coil of 200 turns and area 0.015 m^2 is rotated in a uniform magnetic field of 0.40 T at a constant angular velocity of $50 \text{ rad}\cdot\text{s}^{-1}$. Find the peak emf generated.

$\epsilon_0 = NBA\omega = (200)(0.40)(0.015)(50) = 200 \times 0.40 = 80$; $80 \times 0.015 = 1.2$; $1.2 \times 50 = 60 \text{ V}$.

Answer: $\epsilon_0 = 60 \text{ V}$.

Quick check · multiple choice

A bar magnet is pushed into a coil of wire connected to a sensitive ammeter. According to Lenz's law, the direction of the induced current is such that it:

A Has no particular direction

B Aids the motion of the magnet into the coil

C Opposes the motion of the magnet into the coil

D Is always in the same direction regardless of magnet motion

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A flat circular coil of 150 turns and radius 0.040 m lies with its plane perpendicular to a uniform magnetic field that decreases uniformly from 0.80 T to 0.20 T over a time interval of 0.50 s . [7]
- (a) Calculate the area of the coil.
- (b) Calculate the change in magnetic flux through one turn of the coil.
- (c) Calculate the magnitude of the average emf induced in the coil during this time.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $A = \pi r^2 = \pi(0.040)^2 = 5.027 \times 10^{-3} \text{ m}^2$.

(b) $\Delta\Phi = \Delta B \times A = (0.80 - 0.20)(5.027 \times 10^{-3}) = 0.60 \times 5.027 \times 10^{-3} = 3.016 \times 10^{-3} \text{ Wb}$.

(c) $\varepsilon = N(\Delta\Phi/\Delta t) = 150 \times (3.016 \times 10^{-3}/0.50) = 150 \times 6.032 \times 10^{-3} = 0.905 \text{ V}$.

Quick check answer

C. Lenz's law states the induced current opposes the change causing it, so the induced magnetic field repels the incoming magnet, opposing its motion into the coil, consistent with energy conservation.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

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